

INNOVATIONS IN FINANCE

LECTURE 5 : AI IN FINANCE

Make an impact



INTRODUCTION



Make an impact



LET'S ASK IT SOMETHING SIMPLE

Live. Now. On the screen. Watch carefully — and write the answer down.

THE QUESTION

"A bond pays a 4.35% annual coupon, matures in 7 years, face value €1,000. The market yield is 5.10%. What is its price today?"

A first-year finance exercise. One right answer. A calculator settles it in twenty seconds.

THEN DO THIS

- 1 Write down the answer it gives.
- 2 Open a brand-new conversation.
- 3 Paste exactly the same question.
- 4 Compare.

WHAT WE ARE TESTING — AND IT IS NOT INTELLIGENCE

Is it right?

Check it against a calculator. Do not take the model's word for it.

Is it stable?

Same question, same model, five minutes later. Same answer?

Does it hedge?

Does it warn you it might be wrong? Or is it equally confident either way?

The third question is the important one, and almost nobody asks it.

A calculator that is wrong tells you nothing. A colleague who is unsure says so. This says neither.



IN CASE THE WIFI FAILS: WHAT YOU WOULD HAVE SEEN

The same question, the same model, two separate conversations

CONVERSATION 1

$$P = 43.50 \left[\frac{1 - (1.051)^{-7}}{0.051} \right] + \frac{1,000}{(1.051)^7}$$

$$P \approx \text{€}954.56$$

Answer

Bond price $\approx$ €954.56

Because the market yield (5.10%) is higher than the coupon rate (4.35%), the bond sells below its €1,000 face value. biz.libretexts.org

CONVERSATION 2

• Maturity: $n = 7$ years

The bond-pricing formula is:

$$P = C \left[\frac{1 - (1 + y)^{-n}}{y} \right] + \frac{F}{(1 + y)^n}$$

Substitute:

$$P = 43.50 \left[\frac{1 - (1.051)^{-7}}{0.051} \right] + \frac{1,000}{(1.051)^7}$$

$$P \approx 43.50(5.786) + 708.80$$

$$P \approx \text{€}960.52$$

Answer: €960.52

Because the market yield (5.10%) is higher than the coupon rate (4.35%), the bond trades below its €1,000 face value. statgarden.com

LOOK AT WHAT IS IDENTICAL IN BOTH

The tone. The confidence. The correct financial reasoning around the number. The professional formatting. Everything except the number itself — and nothing anywhere signals which one to trust.

The explanation is right. The number is a guess dressed as a result.

That combination — sound reasoning wrapped around an unreliable figure — is exactly what makes this dangerous in finance.



IT IS NOT CALCULATING. IT NEVER WAS.

The single most useful thing to understand about this technology

WHAT IT ACTUALLY DOES

Given everything so far, produce the most plausible next fragment of text. Then repeat. That is the whole mechanism.

SO WHEN YOU ASK FOR A NUMBER

It is not performing a multiplication and returning the result.

It is producing a sequence of digits that looks like the kind of answer that follows that kind of question.

Sometimes those coincide. Nothing inside the model knows when.

WHICH EXPLAINS EVERYTHING YOU JUST SAW

Fluent reasoning	language is exactly what it was built for
Unstable numbers	there is no calculation to be stable
Total confidence	it has no concept of being unsure
No error signal	wrong and right are produced by the identical process

THE ANALOGY THAT MAKES IT CLICK

It is not a calculator that sometimes slips. It is an extraordinarily well-read colleague who has never used a calculator in their life, answers instantly from memory of things they have read, and has never once said "let me check that".

You would find that colleague useful. You would not let them close the accounts.





WHAT FINANCE NEEDS, AND WHAT THIS GIVES YOU

Line by line, it is almost a perfect mismatch — for one half of the job

FINANCE REQUIRES

Exact

The number is right, or it is not. There is no "approximately" in a payment.

Reproducible

The same input gives the same output, today and in six months.

Traceable

Every figure comes from a source you can point to.

Auditable

A regulator can reconstruct how you got there.

Honest about doubt

A good analyst says "I am not sure".

AN LLM PROVIDES

Plausible

Looks like the kind of number that belongs there.

Non-deterministic

The same question can give different answers.

Unsourced

It cannot tell you where a number came from, because it did not come from anywhere.

Opaque

"The model produced it" is not a reconstruction.

Uniformly confident

It has no mechanism for expressing uncertainty about its own output.

None of this makes it useless. It makes it wrong for one specific half of the job — and that half happens to be the numbers.





THE RULE TO TAKE OUT OF TODAY

If you remember one sentence from this session, make it this one

Never let a model produce a number.

Let it produce the code, the query or the classification that produces the number.

WRONG

"What is the IRR of these cash flows?"

You get a number. You cannot check where it came from, and it may change tomorrow.

RIGHT

"Write me the formula, or the Python, that computes the IRR of these cash flows."

You get something you can read, test, re-run and hand to an auditor.

WHY THIS IS NOT A WORKAROUND BUT THE ACTUAL METHOD

The model is excellent at the thing it was built for — turning your intention into language, including the language called code. Code is deterministic, readable, testable and auditable. So you use the model for what it is good at, and you let a machine that actually computes do the computing. This is exactly what session 8 will have you doing.





WHERE WE ARE GOING

And no, the rest of this session is not an argument against AI

1 What it is

Four generations of AI, and why the newest is not always the right one.

2 Where it works

Credit, fraud, trading, advice, text, the finance function, agents — with honest verdicts on each.

3 Who is responsible

The block that decides whether any of this is usable in a regulated business.

4 What breaks

Model risk, hallucination, feedback loops, bias.

5 What the law says

The EU AI Act — and the deadline that moved two months ago.

6 Your job

The honest answer to the question you actually came in with.

Session 4 gave you the raw material. This session is the engine that turns it into decisions — and everything wrong with the data comes out faster here.





AI IN FINANCE



Make an impact



EARLY DAYS: SYMBOLIC AI & EXPERT SYSTEMS (1960S–1980S)

Generation 1 — a human writes every rule

THE CONCEPT

Computers mimicking human reasoning with explicit rules: “if this, then that”.

APPLICATIONS IN FINANCE

- Early credit scoring models.
- Rule-based fraud detection — e.g. “flag if transaction > \$10,000”.

LIMITS

- Rigid.
- Needed constant manual updating.
- Could not handle complexity — or anything nobody had anticipated.

EXAMPLE: OPERATION CATEGORISATION

IF the label contains “SNCF” **THEN** Travel. **IF** it contains “URSSAF” **THEN** Social charges.

Fast, free and perfectly explainable. And somebody has to write — and maintain — the next ten thousand rules.





THE RISE OF MACHINE LEARNING (1990S–2000S)

Generation 2 — the machine derives the rules from examples

THE CONCEPT

Instead of programming rules, feed data → the model learns the patterns.

TECHNIQUES

- Decision trees
- Regression
- Support vector machines

FINANCE APPLICATIONS

- Risk modelling & Value-at-Risk
- Algo trading (pattern recognition)
- Credit scoring using more variables

IMPACT

Better at handling large datasets.
But still limited by computing power.

EXAMPLE: OPERATION CATEGORISATION

Trained on thousands of labels already categorised by humans, the model learns that this kind of label, at this kind of amount, usually means “Supplies” — and returns a confidence score.

Nobody wrote the rule. And you can measure it: hold back data it never saw, count the errors.





BIG DATA + DEEP LEARNING REVOLUTION (2010S)

Generation 3 — many more layers, made possible by data and GPUs

THE CONCEPT

Neural networks inspired by the brain: layered models. Early layers detect simple patterns, later layers combine them.

WHY NOW?

Explosion of data — social media, mobile banking, transactions.

Stronger computing — GPUs make training large networks affordable.

FINANCE APPLICATIONS

- High-frequency trading strategies.
- Chatbots & virtual assistants in banking.
- Image/text recognition for compliance — e.g. document verification in KYC.

Key event — 2012: the “ImageNet breakthrough”.

Deep learning beats every other approach at recognising images, and the hype spreads into finance.





GENERATIVE AI ERA (2020S–)

Generation 4 — from prediction to creation

THE CONCEPT

Large Language Models (LLMs) like GPT generate text, code and images. The difference: from prediction to creation.

FINANCE APPLICATIONS

- Report writing, portfolio commentary, compliance summaries.
- Conversational banking assistants (personalised financial advice).
- Scenario generation for risk management.
- Code generation for faster model development.

RISKS

- Hallucinations
- Bias
- Explainability
- Regulatory scrutiny

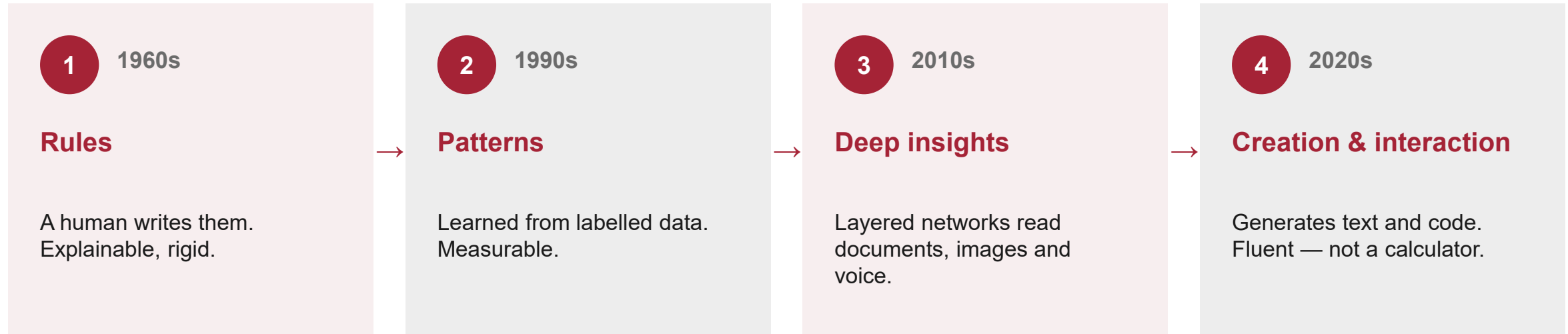
EXAMPLE: OPERATION CATEGORISATION

Give it the label and ask. It reads “SARL DUP” the way a human would and guesses from context — superb on the strange label nobody has seen, but slower, dearer and not deterministic.



WRAP-UP: EVOLUTION IN ONE SENTENCE

Four generations, each one expanding what AI does in finance



From rules → to patterns → to deep insights → to creation & interaction.

Each stage expanded the role of AI in finance — from back-office number crunching to front-office decision-making.



LLM TRAINING & FINANCE: GARBAGE IN, GARBAGE OUT

What the model read determines what it writes

1 How LLMs are trained

Massive web content: Wikipedia, blogs, forums, news, social media.

Learns patterns → predicts the next word.

2 The problem in finance

Online finance content is often oversimplified, biased or promotional — or simply wrong.

⇒ Risk of hallucinations and low-quality answers.

3 Portfolio commentary

A model trained on poor data produces exaggerated, incorrect or invented narratives.

Example: “The dollar rose because of oil prices” — with no evidence.

SOLUTIONS

Fine-tuning — train further on reliable financial datasets (Bloomberg, in-house research).

RAG (Retrieval-Augmented Generation) — ground every answer in trusted documents you control. We come back to this in “Give it a library”.





DEFINING AI IN FINANCIAL SERVICES

The vocabulary you need for the rest of the session

WHAT IS AI?

Artificial Intelligence: machines that mimic human intelligence — learning, reasoning, problem-solving.

Supervised learning

Labelled data → used in credit scoring and fraud detection.

Unsupervised learning

Finds patterns in unlabelled data → customer segmentation, anomaly detection.

Machine vs deep learning

ML learns from structured data (suspicious card activity). DL uses layered neural networks (financial documents, voice).

Natural language processing

Interprets and generates human language → support chatbots, market sentiment analysis.

KEY USE CASES

Fraud alerts

Real-time anomaly detection using ML.

Robo-advisors

Portfolio allocation powered by AI.

Chatbots

24/7 support using NLP and ML.





TWO VERY DIFFERENT THINGS CALLED "AI"

Confusing them is the most common and most expensive mistake in the field

A MACHINE-LEARNING MODEL

A LARGE LANGUAGE MODEL

Trained on

Your data. Your defaults, your fraud cases, your customers.

The internet. Everyone's text, up to a cut-off date.

Trained for

One task. Predict this specific outcome.

Any task involving language. No task in particular.

Can you test it?

Yes. Hold back data, measure accuracy, compare to a baseline.

Not in the same sense. There is no accuracy score for "write me a summary".

Output

A probability. 0.83 that this application defaults.

Text. Which may contain a probability it made up.

Is it stable?

Yes, until you retrain it. Same input, same output.

No. Same input, possibly different output.

Good for

Scoring, ranking, detecting, forecasting — at scale.

Reading, summarising, extracting, drafting, classifying, coding.

When a vendor says "we use AI", your first question is which of these two — and your second is what they tested it against.





ONE PROBLEM, THREE TECHNOLOGIES, THREE DECADES

Disclosure — this is KowiKan, my company, and we use all three at once

The problem: "CB SARL DUP*** 06/03 -428,50". Which expense category is that?

Rules

1960s – expert systems

If the label contains SNCF, it is travel. If it contains URSSAF, it is social charges.

Fast, free, perfectly explainable. Breaks on anything unforeseen, and somebody has to maintain thousands of rules.

Machine learning

1990s – supervised models

Trained on millions of labels already categorised by humans. Learns that this pattern usually means that category.

Handles volume and variation, gives a confidence score, and is testable. Needs a large labelled history to start.

Large language model

2020s – generative

Give it the label and ask. It reads "SARL DUP" the way a human would and guesses from context.

Superb on the rare, strange, never-seen-before label. Slower, more expensive per line, and not deterministic.

In production we run all three. Rules take the easy majority for nothing, the ML model takes the bulk, and the LLM handles the residue no one has ever seen.





AI-DRIVEN CREDIT SCORING

From a few fixed rules to thousands of signals

TRADITIONAL CREDIT SCORING

- Based on limited financial history — income, debt ratio, credit history.
- Fixed rules and linear models (e.g. FICO score).
- Binary decision-making: approve / reject.
- Often excludes thin-file or unbanked customers.

AI-BASED CREDIT SCORING

- Uses machine learning to analyse large, complex datasets.
- Continuously learns from past defaults, repayment patterns and behaviour.
- More granular, probabilistic risk predictions — adapted in real time.

ALTERNATIVE DATA IN AI MODELS

- **Behavioural:** payment patterns, app usage, browsing.
- **Social:** online presence, peer associations (controversial).
- **Transactional:** bank and mobile-wallet activity (especially in emerging markets).

BENEFIT

More inclusive — especially for gig workers and unbanked populations.

CHALLENGE

Raises concerns about bias, explainability and data privacy.

ZEST^{AI}

 Upstart



CASE STUDY – UPSTART’S AI CREDIT SCORING MODEL

A US fintech that sells its credit model to banks

WHO IS UPSTART?

US fintech founded in 2012. Uses AI/ML to assess creditworthiness and partners with banks, which lend on its risk assessment.

HOW IT SCORES DIFFERENTLY

- Considers more than 1,000 variables: education (college, major, GPA), employment history, income stability, digital footprint, behavioural signals.
- Supervised machine learning models.
- Continuously retrained on repayment performance.

43%

Higher approvals

Better accuracy lets bank partners approve more applicants.

33%

Lower APR

...or offer lower rates at the same approval rate.

92%

Fully automated

Instant answer to a loan request, 24/7.

Figures as reported by Upstart.

Challenges & risks

Regulatory attention on fairness and explainability · scrutiny of bias in non-traditional inputs · transparency for consumers.





THE TRADE-OFF NOBODY CAN ENGINEER AWAY

The better the credit model performs, the less you can explain it

"Your application was refused." Under EU law, that is not an explanation.

A scorecard, 10 variables

Modest accuracy

"Refused because your debt-to-income ratio is 47%, above our 40% limit." A human can say that out loud, and the applicant can act on it.

A gradient-boosted model, 200 variables

Better accuracy

You can list which variables mattered most. You cannot really say why this person, in a way they would accept.

A deep model, 1,000+ variables

Best accuracy

You can produce an approximation of an explanation. It is a reconstruction after the fact, not the reason.

THE COMMERCIAL COST OF THE TRADE-OFF

A less accurate model refuses good customers and accepts bad ones. That difference is real money, every year.

AND THE REGULATORY ANSWER

The law is quietly pushing regulated lending back toward simpler models. Not by banning anything — by requiring an explanation.





REAL-TIME FRAUD PREVENTION WITH AI

From static thresholds to a risk score on every transaction

TRADITIONAL FRAUD DETECTION

- Rule-based systems.
- Static thresholds (e.g. transactions > \$5,000 flagged).
- Delayed alerts — often after the transaction.
- High false positives & poor user experience.
- Hard to adapt to evolving fraud techniques.

AI-DRIVEN FRAUD DETECTION

- Real-time transaction monitoring.
- ML models identify subtle patterns.
- Anomaly detection (behavioural or network-based).
- Dynamic risk score per transaction.
- Continuously improves with new data — fewer false positives.

USE CASE EXAMPLES

Pattern: the user usually shops in Paris; suddenly a charge from Singapore — flagged instantly.

Behaviour: device change, login-time shift, typing speed — real-time alerts.





CASE STUDY – HOW STRIPE RADAR DETECTS FRAUD

Machine learning built into the payment flow itself

WHAT IS STRIPE RADAR?

An ML-powered fraud tool built into the Stripe payment platform.

Blocks fraudulent transactions in real time while minimising false positives.

Used by millions of businesses worldwide.

HOW IT WORKS

- 1 Model trained on billions of transactions.
- 2 Analyses 1,000+ signals: IP address, device fingerprint, purchase history, email reputation...
- 3 Behavioural patterns: time of purchase, velocity of transactions.
- 4 Real-time decision: accept, block or flag.
- 5 Custom rules: merchants add their own.

89%

**of high-risk payments
blocked**

As reported by Stripe.

KEY OUTCOMES

Fewer chargebacks and fraud losses · better conversion, because fewer genuine payments are declined · used by Amazon, Shopify, Slack.





WHY FRAUD DETECTION IS THE BEST CASE IN THIS SESSION

And the four criteria that tell you where else AI will work

1

The feedback loop is fast

A fraudulent transaction is disputed within days. The model learns from its own mistakes almost immediately.

Credit default takes years. Trading signals may never resolve cleanly.

2

Errors are cheap and reversible

A blocked genuine payment is annoying. It is not a life event, and it is undone in a phone call.

A refused mortgage is a life event. A wrong trade is a loss.

3

There is enormous labelled data

Billions of transactions, and every one is eventually labelled fraud or not fraud.

Most finance problems have very few labelled examples of the thing you care about.

4

Nobody demands an explanation

The card was blocked. Neither the customer nor the regulator asks for the model's reasoning.

A credit refusal must be explained. That changes which models you can use.

Score any AI project against these four before anyone writes a line of code.





THE 0.1% PROBLEM

Why accuracy is the wrong way to judge a fraud model

A model that answers “not fraud” to every single payment is 99.9% accurate — and catches nothing.

THE TWO QUESTIONS THAT MATTER

Precision — of the payments I blocked, how many were really fraud?

Get it wrong: genuine customers declined, lost sales.

Recall — of all the fraud that happened, how much did I catch?

Get it wrong: losses and chargebacks.

1,000,000 PAYMENTS, 1,000 OF THEM FRAUD

5,000

flagged

900

frauds caught

4,100

genuine customers
blocked

Recall 90%. Precision 18%.

Good or bad? It depends what a blocked customer costs you.

AND IN ANTI-MONEY LAUNDERING, IT IS WORSE

Industry estimates commonly put the share of AML alerts that turn out to be false positives above 90%. Every one is reviewed by a human — that queue, not the model, is the real cost.





ALGORITHMIC TRADING: WHAT THE MACHINE ACTUALLY DOES

Three different jobs, routinely confused, with very different track records

Execution

How do I buy 400,000 shares without moving the price against myself?

Slice the order across time and venues, adapt to liquidity. This works, it is measurable, and it is where the reliable value is.

Verdict: genuinely solved. Almost all institutional flow is executed this way.

Signal

What should I buy, and when?

Find patterns in prices, flows, fundamentals or alternative data that predict returns.

Verdict: extremely hard. This is where most of the money and most of the disappointment live.

Risk & hedging

What am I exposed to right now, and what happens if it moves?

Continuous portfolio monitoring, scenario generation, automatic hedge adjustment.

Verdict: works well, and it is the least discussed of the three.

When someone says "AI trading", ask which of the three. Two of them work well and nobody writes articles about them.





WHY PREDICTING MARKETS IS HARDER THAN READING X-RAYS

Four reasons, and the fourth has no equivalent anywhere else in AI

1

The signal is tiny

A model that is right 52% of the time is outstanding. In most AI applications, 52% is worthless.

2

The rules keep changing

A cat looks like a cat forever. What worked in markets in 2019 may be meaningless in 2026. Your training data expires.

3

You will find patterns that are not there

Test enough strategies on enough history and some will look brilliant by chance. Backtests are where hope goes to be confirmed.

4

The market adapts to you

This one is unique to finance. An X-ray does not change because you analysed it. A profitable strategy attracts imitators, and the edge is competed away — sometimes by your own execution.

Every other field of AI works on a world that does not fight back. Finance is the exception.





HOW TO DISCOVER A STRATEGY THAT DOES NOT EXIST

Reason 3 from the previous slide, in numbers

THE EXPERIMENT

- 1 Generate 1,000 random trading strategies. Pure noise — no edge at all.
- 2 Backtest each one on ten years of history.
- 3 Keep the ones that are statistically significant at 5%.
- 4 **About 50 survive — each with a beautiful equity curve.**

WHY IT FOOLS PROFESSIONALS

Nobody reports the 950 that failed. The paper, the pitch deck and the fund launch show only the survivors.

The more strategies you test, the higher the Sharpe ratio you should expect from luck alone. Bailey & López de Prado's "deflated Sharpe ratio" corrects for exactly this.

THE DEFENCES

Count your trials

Report how many strategies you tried, not just the winner.

Hold data back

Keep a period you never touch until the very end — and use it once.

Demand a story

Why should this edge exist, and why has nobody competed it away?

Machine learning makes this worse, not better: it can test a million strategies before lunch.





FROM HUMAN ADVISORS TO ROBO-ADVISORS

The same job — automated, and priced for everyone

	TRADITIONAL ADVISORY	AI-POWERED ROBO-ADVISOR
Risk profiling	Interviews and forms, by hand.	Online questionnaire + pattern analysis.
Rebalancing	Periodic, done manually.	Continuous and algorithmic.
Tax	Planned yearly, or on request.	Automated tax-loss harvesting, routinely.
Cost	High fees and minimum investment.	Low or no minimum, reduced fees.
Personalisation	Personal, but hard to scale.	Standardised personalisation at scale.
Access	Scheduled consultations.	24/7, through apps and dashboards.

AI democratises wealth management.
Professional-grade investment services, available to everyday investors.





CASE STUDY – HOW BETTERMENT AUTOMATES INVESTMENT

Modern Portfolio Theory, run by software

1 Risk profiling

A quick questionnaire on time horizon, goals and risk tolerance → a personal risk score (0–100).

2 Portfolio selection

Based on Modern Portfolio Theory. Diversified ETFs — US and international stocks, bonds. ESG options.

3 Automatic rebalancing

Monitors drift daily; buys and sells to hold the target — including after deposits, withdrawals, dividends.

4 Low & transparent fees

0.25% a year for the digital plan, no minimum. 0.65% premium with human advisors. No trading fees.

KEY OUTCOMES

Consistent returns aligned with each client's risk level.

Better investor behaviour — e.g. less panic selling.

Tax efficiency via Tax-Loss Harvesting+ (eligible accounts).



WHAT THE ROBOT ACTUALLY OPTIMISES

Under the app, it is Markowitz — with Markowitz's problems

THE ENGINE

Mean-variance optimisation (Markowitz, 1952): the highest expected return for a given level of risk.

THE INPUT PROBLEM

The optimiser needs expected returns — and nobody knows them.

Small errors in the inputs produce extreme, unstable portfolios. Michaud called optimisers “error maximisers”, which is why robo-advisors constrain them heavily.

THE QUESTIONNAIRE PROBLEM

Risk tolerance is measured with a few questions, on a calm day.

It records what clients say they will do — not what they do when the market falls 30%.

WHERE THE VALUE REALLY IS

Not in the optimisation. In the discipline:

automatic rebalancing, no panic selling, tax-loss harvesting. The robot's real edge is behavioural.

Markowitz gives you the maths. Kahneman explains why the client needs a robot to stick to it.



READING THE MARKET: NLP BEFORE CHATGPT

Finance was turning text into signals long before LLMs

1

Word lists

2000s

Count positive and negative words. Loughran & McDonald (2011) showed that most “negative” words in a general dictionary — liability, tax, cost — are neutral in a 10-K. Finance needs its own vocabulary.

2

Finance-specific models

2019

FinBERT: a language model fine-tuned on financial text to classify the sentiment of each sentence.

3

Large language models

2023–

Read whole documents, answer questions about them, compare this quarter’s call with the last one.

WHAT IT IS USED FOR

Earnings calls

Tone, hesitation, what management avoids answering.

Central banks

Hawkish or dovish? Word-by-word changes between statements.

News & social media

Detecting events in real time, before prices move.

Filings

What changed in the risk-factor section since last year?

Text is the largest dataset in finance, and it sat unused for decades. This is where language models genuinely belong.



SPECIALIST OR GENERALIST? THE BLOOMBERGGPT LESSON

Proprietary data against scale — and a result that surprised the industry

MARCH 2023 — BLOOMBERG BUILDS ITS OWN

A 50-billion-parameter model, trained on roughly half financial data drawn from decades of Bloomberg's own archives.

On finance benchmarks, it beat open models of comparable size.

THEN THE GENERAL MODELS CAUGHT UP

Within months, independent tests found GPT-4 — with no finance-specific training — matching or beating it on many financial text tasks.

Scale and general reasoning had closed most of the gap.

SO WHAT SHOULD A BANK DO?

1

Build its own model

Very expensive, and overtaken by the next general release.

2

Fine-tune a general model

Cheaper — but it still needs clean, labelled in-house data.

3

Retrieve from its own documents

The model stays general; your data stays yours, current and citable (RAG).

The lasting advantage is not the model. It is the data — and the right to use it.





WHERE GENERATIVE AI GENUINELY EARNS ITS PLACE

The honest list — and it is longer than the sceptics admit

Reading long documents

Extract the twelve covenants from a 200-page credit agreement, with the page reference for each.

Summarising

Turn forty analyst notes into a briefing. Then read the originals for the ones that matter.

Drafting

First version of a memo, an email, a commentary. You edit; you do not ship it raw.

Classifying

Is this transaction label a supplier or a salary? Is this clause unusual?

Writing code

The rule from the start of the session. This is the highest-value use in finance, and it is session 8.

Explaining

"Explain this instrument to me as if I were the client." Genuinely excellent, and immediately useful.

AND THE MATCHING LIST OF WHAT IT MUST NOT DO ALONE

- | | | | |
|---------------------------|------------------------------|--------------------------|-----------------------------|
| ● Produce a number | you cannot check | ● Give advice | to a client, unsupervised |
| ● Cite a source | you have not opened yourself | ● Make a decision | someone will have to defend |





THE QUIET REVOLUTION: AI IN THE FINANCE FUNCTION

Less glamorous than trading — and where much of the value is being captured today

TASK	WHAT AI DOES	VERDICT
Cash forecasting	Predicts when each invoice will actually be paid, from each customer's payment history — not from the due date.	<i>Works. And measurable against what actually arrived.</i>
Bank reconciliation	Matches payments to invoices despite partial payments, typos and grouped transfers.	<i>Works. Rules and ML together, humans on the residue.</i>
Invoice processing	Reads supplier invoices, extracts amounts, VAT and due dates, proposes the booking.	<i>Works — if every extracted figure is checked against the source.</i>
Close commentary	Drafts the variance commentary for the monthly close.	<i>A useful draft. The numbers come from the ledger, never from the model.</i>

Same rule as the start of the session: the model classifies and drafts. The figures come from the system of record.





FROM CHATBOT TO AGENT: WHEN THE MODEL CAN ACT

The shift of 2025–2026: models that do things, not just say things

WHAT MAKES AN AGENT

1 A goal

“Pay this month’s approved supplier invoices.”

2 Tools

Access to the ERP, the bank API, the mailbox.

3 A loop

Plan, act, check the result, act again.

4 Permissions

What it may do without asking anyone.

A chatbot answers your question. You act.

An agent takes the action. You review — perhaps.

ALREADY HAPPENING IN PAYMENTS In 2025 Visa (Intelligent Commerce) and Mastercard (Agent Pay) announced frameworks letting AI agents buy on a user’s behalf, with tokenised credentials and spending limits.

A chatbot that is wrong gives you a bad answer. An agent that is wrong gives you a bad transaction.





AN AGENT JUST PAID THE WRONG SUPPLIER. WHO PAYS?

The question the next section is built around

The user	gave the mandate.	<i>How precise was “pay the approved invoices”?</i>
The agent provider	built the model that misread the invoice.	<i>Its contract almost certainly excludes this.</i>
The bank	executed a correctly authenticated payment.	<i>From its side, nothing went wrong.</i>
The supplier	received money it was not owed.	<i>Recovery is possible — slow, and not guaranteed.</i>

WHY THE RULES STRAIN

Payment law assumes a person authorises each payment — strong customer authentication is built around a human proving who they are. An agent acting for you fits none of the boxes cleanly.

WHAT FIRMS DO MEANWHILE

Spending limits · approved-supplier lists · human approval above a threshold · a complete log of every action the agent took.

Delegating the task does not delegate the responsibility.

Hold that thought — it is the next section.





WHO IS RESPONSIBLE?

THE BLOCK THAT DECIDES WHETHER ANY OF THIS IS USABLE



Make an impact



A MACHINE CANNOT BE HELD ACCOUNTABLE

Not as a philosophical point. As a practical one, with a name on a document.

THE TEST

When the advice turns out to be wrong, who loses something? A licence, a job, a reputation, a fee?

The model

Nothing. It cannot be fined, struck off, sued or embarrassed. It has nothing to lose, which is precisely why it cannot be accountable.

The firm

Everything. In regulated advice the licensed entity carries the obligation regardless of what tool produced the recommendation.

The adviser

Their licence and their name. In France a CIF is registered, supervised and personally on the hook for the suitability of the advice.

The client

Their money — and a right to redress that only exists because somebody identifiable is responsible.

"The algorithm decided" has never once worked as a defence — not in front of a regulator, and not in front of a client.





AND CLIENTS DO NOT WANT A MACHINE ANYWAY

The behavioural half of the argument, which is at least as strong as the legal one

WHAT PEOPLE SAY THEY WANT

Low fees. Speed. No appointment. No condescension. A clean app.

All of which a robo-adviser delivers better and cheaper than any human ever could — and that is genuinely why they succeeded.

WHAT THEY WANT AT 9AM ON A BAD MONDAY

The market is down 18%. Their savings are down 18%.

They do not want a dashboard, a push notification, or a well-worded paragraph about long-term horizons. They want to call somebody who knows their name and hear them say it will be alright.

AND THE INDUSTRY ALREADY VOTED ON THIS

The pure robo-advisers added human advisers. Betterment's premium tier costs roughly 0.65% instead of 0.25% — and the difference is access to a person. Clients pay more than double, voluntarily, for the thing the technology was supposed to make unnecessary.

The most valuable thing an adviser does is not build the portfolio. It is stop the client selling it at the bottom — and no notification has ever managed that.





THE MACHINE DOES NOT REMOVE YOUR BIASES. IT ADDS SOME.

Kahneman, meet the algorithm

WHERE ALGORITHMS GENUINELY HELP: NOISE

Two underwriters given the same file disagree far more than their managers expect (Kahneman, Sibony & Sunstein, Noise, 2021). A scoring model — not an LLM — gives the same answer to the same file every time.

1 Automation bias

We accept the machine's output without checking — especially when tired, rushed or junior. The demo at the start of this session relied on it.

2 Algorithm aversion

After seeing an algorithm make one mistake, people abandon it — even when it beats humans (Dietvorst, Simmons & Massey, 2015). We forgive human errors, not machine ones.

3 Anchoring

Show an analyst the model's number first, and their "independent" judgement drifts towards it. The second opinion is no longer independent.

Good practice: form your own view before you look at the model's. Then compare.





SO WHAT DOES THE JOB ACTUALLY LOOK LIKE?

The division of labour that is emerging — and where you fit into it

THE MACHINE DOES

the volume

Screening thousands of files · monitoring portfolios continuously · drafting the first version · extracting terms from documents · flagging anomalies · running the calculation

THE HUMAN DOES

the exceptions and the exposure

The unusual case · the judgement call · the conversation · the decision somebody has to sign · the moment the client is frightened

AND CRITICALLY

the human owns the output

Not "the model suggested it". You checked it, you understood it, you signed it. That is the job description, and it does not disappear.

WHICH CHANGES WHAT A JUNIOR IS FOR

The traditional junior job was to produce the first draft, gather the data and format the deck. A machine now does all three faster. The junior job that remains is to check the machine — which requires knowing enough to spot when it is wrong.

That is a harder job than the one it replaced, and it is why you still need to learn to price a bond by hand.





WHAT BREAKS

FOUR FAILURE MODES, AND NONE OF THEM ARE BUGS



Make an impact



MODEL RISK: THE MAP IS NOT THE TERRITORY

We had this conversation in session 3. Here it is again, with more compute.

1998 — LONG-TERM CAPITAL MANAGEMENT

Two of the men behind the option pricing model sat on the board. One had received the Nobel Prize the year before.

The mathematics was not wrong. The world stopped matching the assumptions, and correlations that were independent all moved together.

2026 — THE SAME STRUCTURE

A credit model trained on 2015–2024 data has never seen a rate shock, a pandemic-scale disruption or a regime change.

It will keep producing confident scores through all three. Nothing in the output will indicate that the world has left the training set.

THE THREE QUESTIONS A MODEL RISK FUNCTION ASKS

What does it assume?

Every model has assumptions. If nobody can list them, nobody understands the model.

When was it last wrong?

A model nobody has seen fail is a model nobody has tested properly.

What would make us stop using it?

Define the failure condition in advance. Afterwards, everyone finds a reason to keep going.

A model is a simplification with assumptions attached. It works until the world stops matching them — and the model is the last thing that notices.





MODEL RISK MANAGEMENT: WHO CHECKS THE MODEL?

Banks solved part of this problem years ago — and LLMs break part of the solution

THREE LINES OF DEFENCE

1 The developers

Build the model, document its assumptions, test it.

2 Independent validation

A separate team whose job is to break it: “effective challenge” (US Federal Reserve, SR 11-7, 2011).

3 Internal audit

Checks that the process was actually followed.

WHAT STILL WORKS FOR MACHINE LEARNING

- An inventory of every model in use.
- Validation before first use.
- Monitoring for performance drift.
- A retirement condition, set in advance.

WHAT LLMS BREAK

- The vendor updates the model without telling you.
- The prompt is part of the model.
- The same input can give different outputs.
- You cannot validate a version that changes under you.

If you cannot freeze it, document it and re-run it, you cannot validate it — and a bank cannot use it for a decision.





HALLUCINATION IS NOT A BUG YOU CAN REPORT

It is the same process that produces correct answers, with a different outcome

MATA V. AVIANCA, NEW YORK, 2023

Two lawyers filed a brief citing six judicial decisions that did not exist. ChatGPT had produced them, complete with docket numbers, quotations and internal citations.

When challenged, one of them asked the model whether the cases were real. It said yes. The court sanctioned them.

WHY THIS IS STRUCTURAL, NOT CARELESS

The model was not lying and was not malfunctioning. Asked for cases supporting a position, it produced text of the shape that citations take.

A real citation and an invented one are generated by exactly the same mechanism. There is nothing to detect.

NOW TRANSPOSE IT INTO FINANCE

- **An invented figure** in a client report, correctly formatted, in the right currency, of the right magnitude.
- **An invented regulation** cited with an article number, in a compliance memo nobody re-checks.
- **An invented fact** about a company, in an investment note that goes to a committee.

Session 1 asked you to check where the Rothschild story came from. Same discipline, new machine — except this one invents the source as well as the story.





THE MODEL CREATES THE DATA THAT PROVES IT RIGHT

The failure mode nobody teaches, and the hardest one to see from inside

You reject an application. You will never find out whether they would have repaid.

- 1 The model refuses a profile** because past applicants like this defaulted.
- 2 Those applicants are never funded** so no repayment data is ever created for them.
- 3 The model retrains** on accepted applicants only — a population it selected itself.
- 4 It becomes more confident** about a belief that can no longer be tested. The error is now permanent.

IT HAS A NAME

Selection bias, made self-reinforcing. Lenders spend real money on "reject inference" trying to correct for it.

AND IT COMPOUNDS WITH BIAS

If the model under-serves a group, that group generates no counter-evidence. The prejudice becomes the data.





WHEN EVERYONE USES THE SAME MODEL

The risks that only appear at the level of the whole market

1 Herding

If most firms train similar models on similar data, they will want to sell the same assets at the same moment. Diversity of opinion is what keeps markets liquid.

Gensler & Bailey, 2020

2 Concentration

A handful of cloud providers and model vendors sit under most of the industry. One outage or one flawed update, and thousands of institutions fail together.

Financial Stability Board, 2024

3 Speed

6 May 2010: the Dow Jones fell nearly 1,000 points in minutes, then recovered most of it within the hour. Automated systems reacted to each other faster than any human could step in.

The “Flash Crash”

WHY NO RISK MODEL CAN SEE IT

Each firm’s model can be individually sound while the system becomes fragile. Model risk is assessed firm by firm; systemic risk sits in nobody’s model.

Session 3, again: things that were independent all move together — only faster.





SO HOW DO YOU ACTUALLY BUILD THIS?

TWO FAILURES, TWO ARCHITECTURES



Make an impact



TWO FAILURES, TWO FIXES

Neither makes the model smarter. Both change what it is allowed to do.

It cannot calculate

Because it predicts text, and a number is text to it.

Give it a calculator

Tool use. The model calls real code, which computes, and hands back an exact result.

It cannot source

Because it generates facts rather than retrieving them — including the citations.

Give it a library

Retrieval. The model is handed the relevant passages first, and answers only from those.

Notice what both have in common: they take work away from the model. That is the design principle, and it is the opposite of what most people assume.





GIVE IT A CALCULATOR: TOOLS AND MCP

The model decides WHAT to compute. Real code decides the answer.



WHAT MCP ACTUALLY IS

A standard way to expose a tool so any model can call it — instead of every model needing a bespoke integration with every tool.

WHERE YOU HAVE SEEN THIS BEFORE

Session 2. PSD2 forced one standard interface onto bank data, and an industry appeared. A standard plug turns integrations into a market.

The number never passes through the probabilistic half of the diagram. That is the entire trick.



GIVE IT A LIBRARY: RETRIEVAL

Stop asking it what it remembers. Hand it the document and ask what it says.

1

The question

"What does our policy say about onboarding a client from a high-risk country?"

2

Search first

The system searches a trusted store — your policies, the regulation, your research — and pulls the relevant passages.

3

The passages go in

They are placed into the prompt alongside the question. The model has not answered anything yet.

4

Answer from those, and cite

The model summarises what the passages say, with a reference you can click.

WHAT CHANGES

The model moves from **GENERATING** facts to **FINDING** them.

And you get a citation that points at a document you own — which is the answer to the Avianca case.

AND WHAT DOES NOT

- If retrieval misses, it still answers — from memory
- Bad documents in, confident wrong answers out
- It does not fix arithmetic
- A citation proves the passage exists, not that it was read correctly

WHICH IS WHY SESSION 4 CAME FIRST

Retrieval is only as good as the document store behind it. Governance, freshness and quality of that store are now your problem — a data problem wearing an AI costume.

THE RULE, RESTATED

**Never cite a source you have not opened.
Retrieval gives you one you can.**





AND SOMETIMES THE RIGHT TOOL IS A SPREADSHEET

The cross-check for the opening demo — open it, it is on the course page

WHAT THE WORKBOOK DOES

- Every cash flow, every discount factor, one row per period
- Three independent routes to the same price
- Your two LLM answers typed in, and the error computed
- And that error scaled to a €10m position
- A yield sensitivity table — duration, felt not defined

THE THREE ROUTES AGREE

Sum of discounted cash flows	€956.76
Closed-form annuity formula	€956.76
Excel's own PV function	€956.76

Every time you open the file. Next week, next year, on anyone's machine.

WHY THIS SLIDE EXISTS AT ALL

It is not more intelligent than the model. It is dumber, in every sense — and that is precisely the property you want. It cannot improvise, cannot be persuaded, and cannot have an off day. It shows its working, and its working is the same every time.

The gap between the two model answers was €5.20 on one bond. On a €10 million position, that is roughly €52,000 — for a discrepancy that looked like rounding.





THE EU AI ACT

AND THE DEADLINE THAT MOVED WHILE YOU WERE ON HOLIDAY



Make an impact



EUROPE REGULATED THE USE, NOT THE TECHNOLOGY

Four tiers of risk — and one of them is the first thing you studied today

Unacceptable <i>Banned outright</i>	Social scoring by public authorities, certain manipulative systems.
High risk <i>Heavily regulated</i>	Creditworthiness assessment of individuals. Risk pricing in life and health insurance. Recruitment. Access to essential services.
Limited risk <i>Transparency obligations</i>	Chatbots must say they are machines. AI-generated content must be marked. Deepfakes must be labelled.
Minimal risk <i>Essentially unrestricted</i>	Spam filters, recommendation engines, most business software.

Note where credit scoring sits. The first application you studied this morning is the one Europe decided to regulate most heavily.





AND THEN EUROPE BLINKED

The high-risk deadline moved. This happened two months ago.

Aug 2024

The AI Act enters into force

The world's first comprehensive AI regulation. Applied in phases.

Feb 2025

Prohibitions apply

Banned practices, plus an obligation for staff working with AI to be trained in it.

Aug 2026

Transparency obligations apply — now

Chatbot disclosure, marking of AI-generated content, deepfake labelling. These are live.

Dec 2027

High risk — postponed from August 2026

Under the Digital Omnibus, the heavy regime for Annex III uses — including creditworthiness assessment — was deferred by roughly sixteen months.

Session 4: Europe delayed Copernicus under American pressure. Session 5: Europe delayed its own AI rules under competitiveness pressure. Notice the pattern.





WRAP-UP

THE QUESTION YOU ACTUALLY CAME IN WITH



Make an impact



SORT THESE EIGHT TASKS

Ten minutes in groups of four. Two of them are traps.

THE TASKS

- 1 Summarise 40 analyst notes into a one-page briefing
- 2 Decide whether to grant a €300,000 mortgage
- 3 Extract every covenant from a 200-page credit agreement
- 4 Compute the fair value of a derivatives book for the year-end accounts
- 5 Draft a client email explaining why their portfolio fell
- 6 Flag which of 2 million transactions look fraudulent
- 7 Answer a client's question about their retirement plan, in a chat window
- 8 Write the Python that calculates the portfolio's exposure

THREE BUCKETS

- **Hand it over**
the machine does it, you spot-check
- **With a human check**
the machine drafts, a person verifies and signs
- **Never alone**
a person must do it, or own it completely

Justify each one with the four criteria from the fraud slide.

Then the question that decides it: if this goes wrong, who explains it to whom?

7 min in groups · 3 min debrief · disagreement is expected and is the point.





WHICH FINANCE JOBS CHANGE FIRST

Exposure is about tasks, not job titles

ROLE	WHAT THE MACHINE TAKES	WHAT STAYS HUMAN
Equity research	Reading filings, first drafts of notes, updating the model.	The view, the call with management, owning the recommendation.
Compliance & KYC	Document checks, alert screening, first-level reviews.	Investigations, grey cases, the conversation with the regulator.
Accounting & payables	Data entry, matching, invoice processing.	The close, the estimates, the controls, the auditor.
Private banking	Portfolio reports, market commentary.	The relationship, the frightened client, the life decision.
Trading	Execution, routine hedge adjustments.	Taking risk, client flow, judgement in a crisis.

In every row, what goes is the volume. What stays is the judgement — and someone has to learn it.





SO — WILL YOU BE REPLACED?

The honest answer, from someone who has been doing this for thirty years

WHAT GENUINELY GOES

- Producing the first draft
- Gathering and formatting the data
- Summarising documents nobody had time to read
- Building the deck
- Answering the same question for the hundredth time

That was most of a junior job. Say it plainly.

WHAT DOES NOT

- Deciding, and signing your name to it
- Noticing that the output is wrong
- The conversation when the client is frightened
- Knowing which question to ask in the first place
- Being the person who is accountable

BUT NOTICE THE PROBLEM WITH THAT

The left-hand column is how people used to LEARN the right-hand column. You became good at spotting a wrong number by producing ten thousand of them. If the machine does the ten thousand, how do you become the person who can check it? Nobody has solved that, and it is the real problem with your generation's entry into this industry.

So learn to do it by hand anyway. Not out of nostalgia — because judgement is built from repetition, and you will be paid for judgement.





WHAT TO TAKE OUT OF TODAY

Five things — the first is the rule, the last is the career advice

1

Never let a model produce a number.

Let it produce the code, the query or the classification that produces the number. That rule survives every future model release.

2

"AI" is two different things.

A machine-learning model is trained on your data for one task and is testable. A language model is trained on everything for nothing in particular and is not.

3

Newest is not best.

Rules, ML and LLMs each win at different parts of the same problem. Systems that work in production use all three.

4

A machine cannot be accountable.

It has nothing to lose. In regulated finance, someone must own the output — which is a constraint on design, not a detail.

5

Learn to do it by hand anyway.

Checking is harder than producing. You cannot supervise a machine doing something you never learned to do.

Before next week: find one financial claim produced by an AI, verify it against the primary source, and report what you found.





AI IN FINANCE

You ask a language model for the price of a bond and get €956.20. What have you actually obtained?

- A.** A sequence of digits that resembles the answer to that kind of question.
- B.** A calculation, which you can verify by repeating it.
- C.** An estimate with a known margin of error.
- D.** The correct answer, since the model was trained on finance textbooks.



AI IN FINANCE

Why is fraud detection a better fit for machine learning than credit approval?

- A.** Fraud is simpler to detect than default.
- B.** Credit approval cannot be automated at all.
- C.** There is more money in fraud than in lending.
- D.** Fast feedback, cheap reversible errors, abundant labelled data, and no duty to explain.



AI IN FINANCE

A credit model refuses a category of applicant. Over three years it becomes more confident in that refusal. Why is that troubling?

- A.** Rejected applicants generate no repayment data, so the belief can never be contradicted.
- B.** It is not — increasing confidence means it is learning.
- C.** Confidence scores are always unreliable.
- D.** The model is overfitting to recent data.



AI IN FINANCE

Under the EU AI Act, where does AI-based creditworthiness assessment of individuals sit?

- A.** Minimal risk — it is only a scoring tool.
- B.** Unacceptable — it is banned outright.
- C.** High risk — with the heaviest obligations, now deferred to December 2027.
- D.** Outside the scope of the Act, which covers only generative AI.



Make an impact

